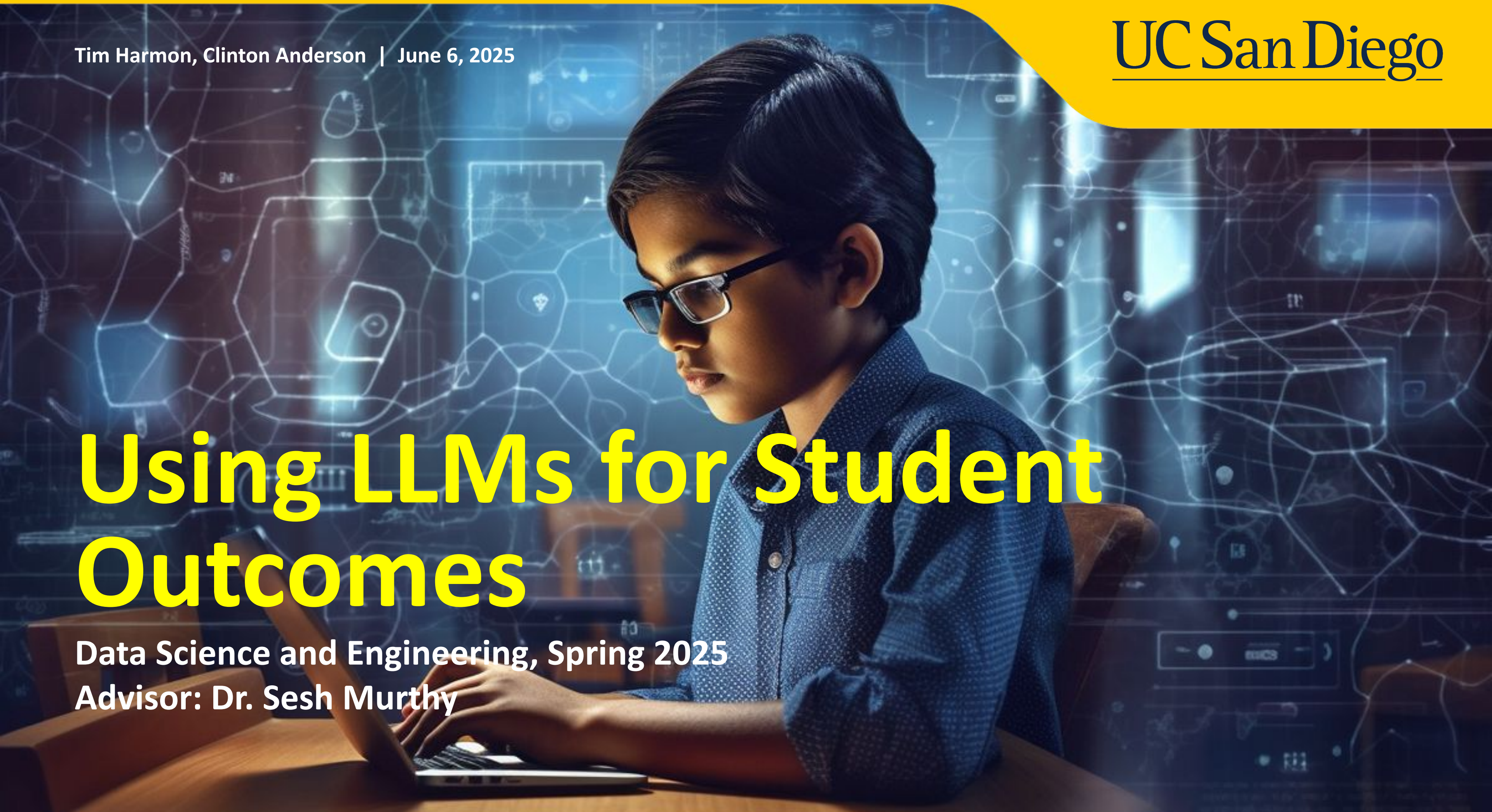


Tim Harmon, Clinton Anderson | June 6, 2025

UC San Diego

Using LLMs for Student Outcomes

Data Science and Engineering, Spring 2025

Advisor: Dr. Sesh Murthy

Overview

INITIAL HYPOTHESES

1. **Performing a hash on limited student data can *sufficiently* reduce exposure and maintain FERPA requirements**

   FAMILY EDUCATIONAL RIGHTS AND PRIVACY ACT

2. **Using synthetic data provides on-demand *statistical evaluation* while preserving initial integrity**

Introduction

UC San Diego

Timothy Harmon

ISC2 CISSP

CISCO CERTIFIED Cybersecurity Associate

Industry Certified

BSides San Diego 2023: Unplugged

CYBERSECURITY INDUSTRY BRAND

Clinton Anderson

English Teacher Grades 1-9 (Japan)

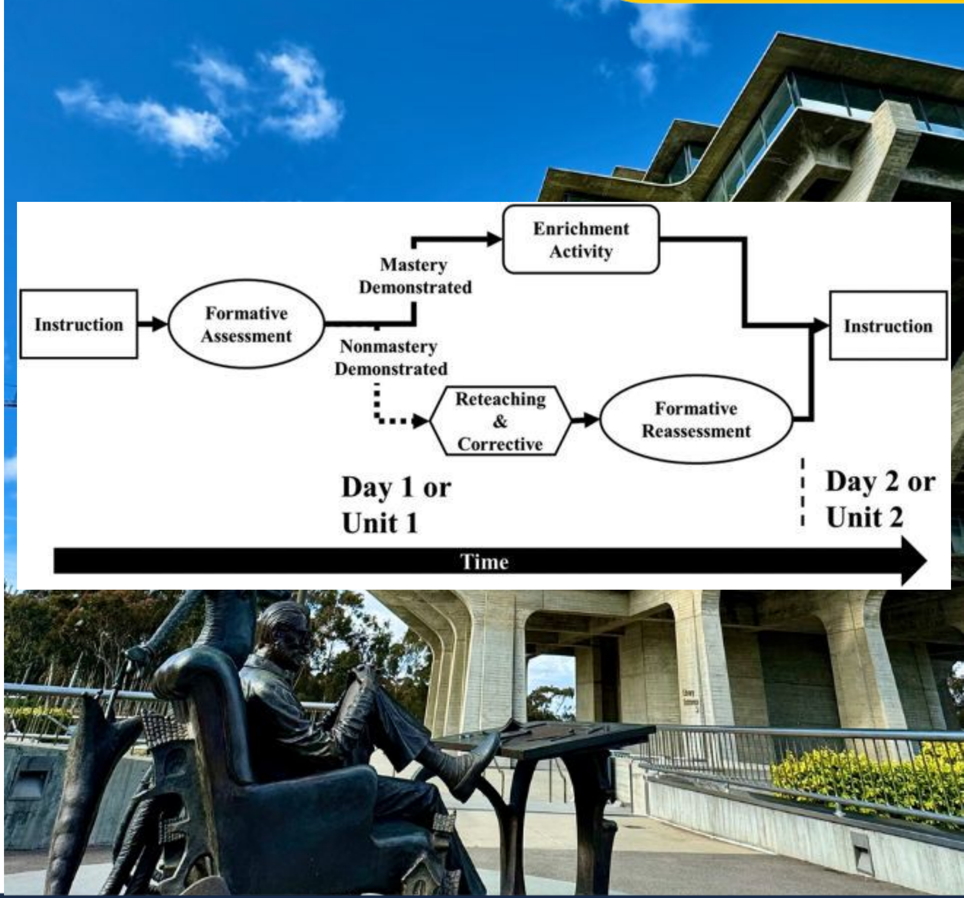
Education Minor

Pasadena City College

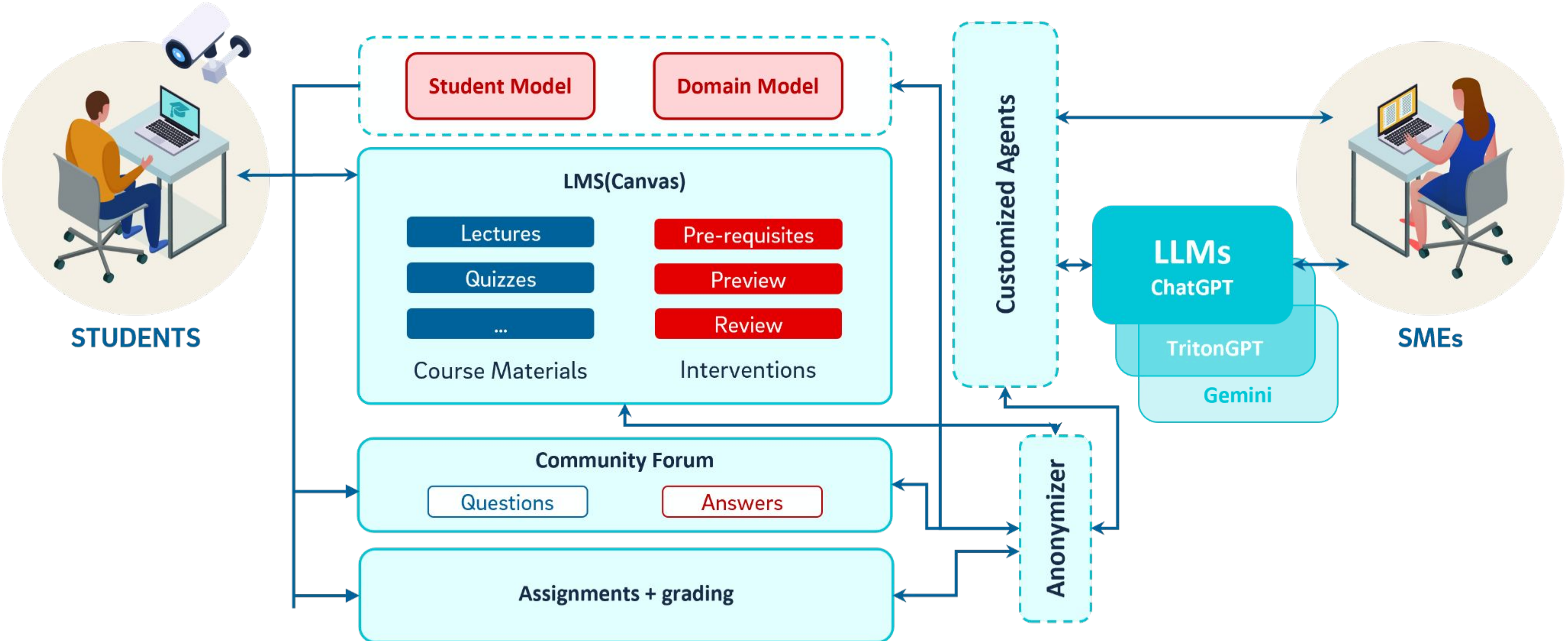
Adjunct Instructor: C++

Problem Definition

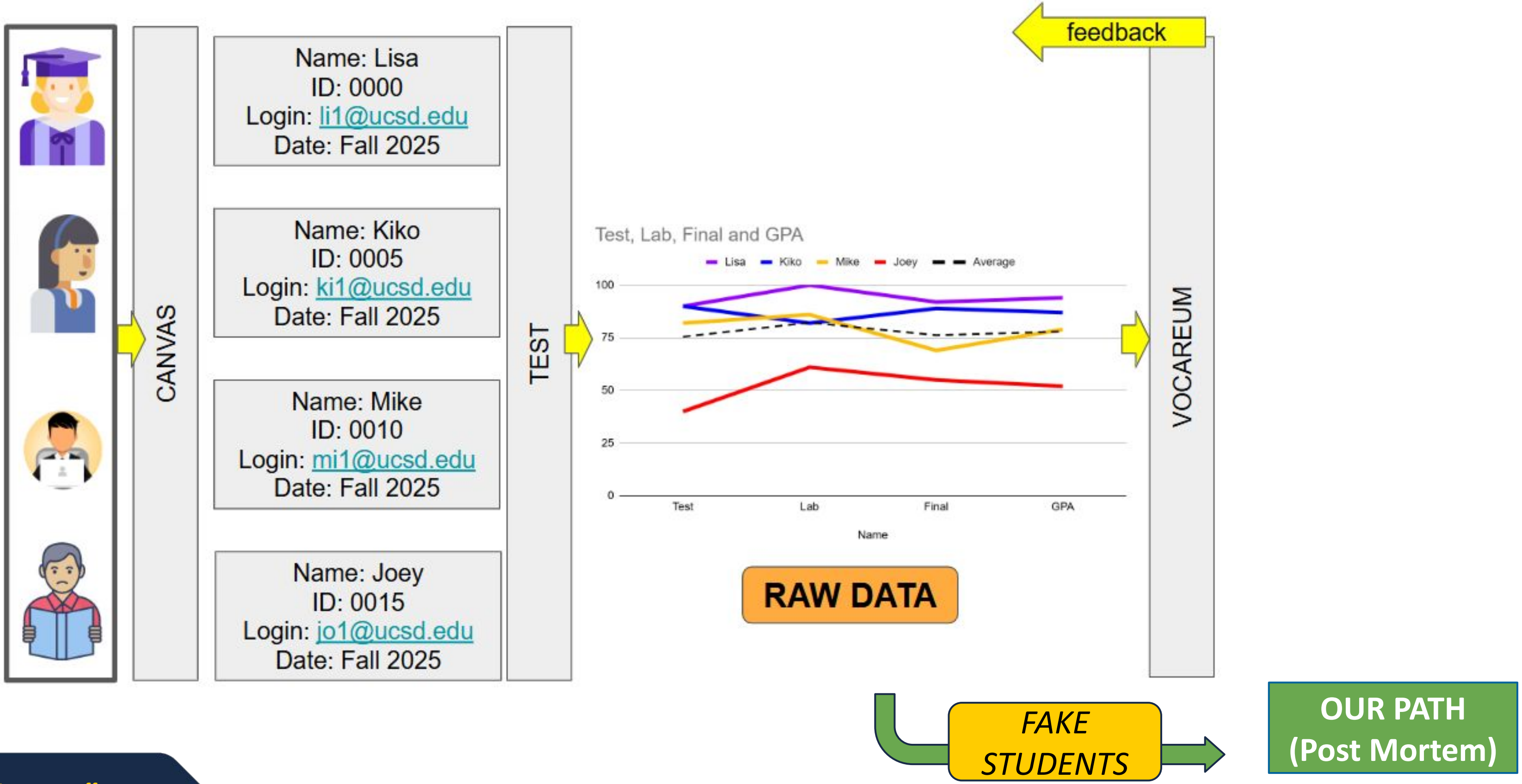
- **Large Language Models (LLMs)**: Personalized assessment and feedback based on students' learning performance.
- By analyzing students' answers, understanding levels, and error patterns during the learning process, LLMs can deliver targeted assessment results and actionable improvement suggestions.
- Detecting / mitigating mistakes **requires specialized expertise**, posing a challenge in their implementation for education.



Approach - Pilot Program ("Hypatia")



Pipeline & Architecture



Initial Attempts - SDV

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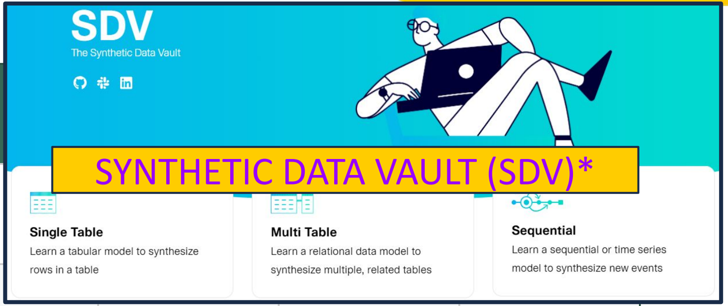


Our Prior “**Raw**” Data

Name	ID	Login	Date	Test			
Lisa	A0000	li@ucsd.edu	Fall 2025				
Kiko	A0001	ki@ucsd.edu	Fall 2025	91	75	92	86

Our Prior ‘**Scripted**’ Data

Name	ID	Login	Date	Test	Lab	Final	GPA
Student18	2b348918419ecc2d	bc@ucsd.edu	Fall 2025	91	75	92	86
Student19	c13c10238d185707	qu@ucsd.edu	Fall 2025	84	89	94	89
Student28	ac0df10ee704d3a41	kb@ucsd.edu	Fall 2025	96	90	96	94

New SDV ‘**Trained**’ Data

	Name	ID	Login	Date	Test	Lab	Final	GPA
0	c49d61	S0048	e5b1eed4@ucsd.edu	Fall 2025	84	76	80	79
1	cd4f10	S0000	54751ce2@ucsd.edu	Fall 2025	85	77	84	93
2	1a251c	S0049	5c2a35e1@ucsd.edu	Fall 2025	83	68	85	77
3	3f3817	S0033	e8479f4c@ucsd.edu	Fall 2025	75	78	95	87
4	c50d19	S0058	39b2b90d@ucsd.edu	Fall 2025	72	59	77	72

Anonymized ‘Login Info’ Here

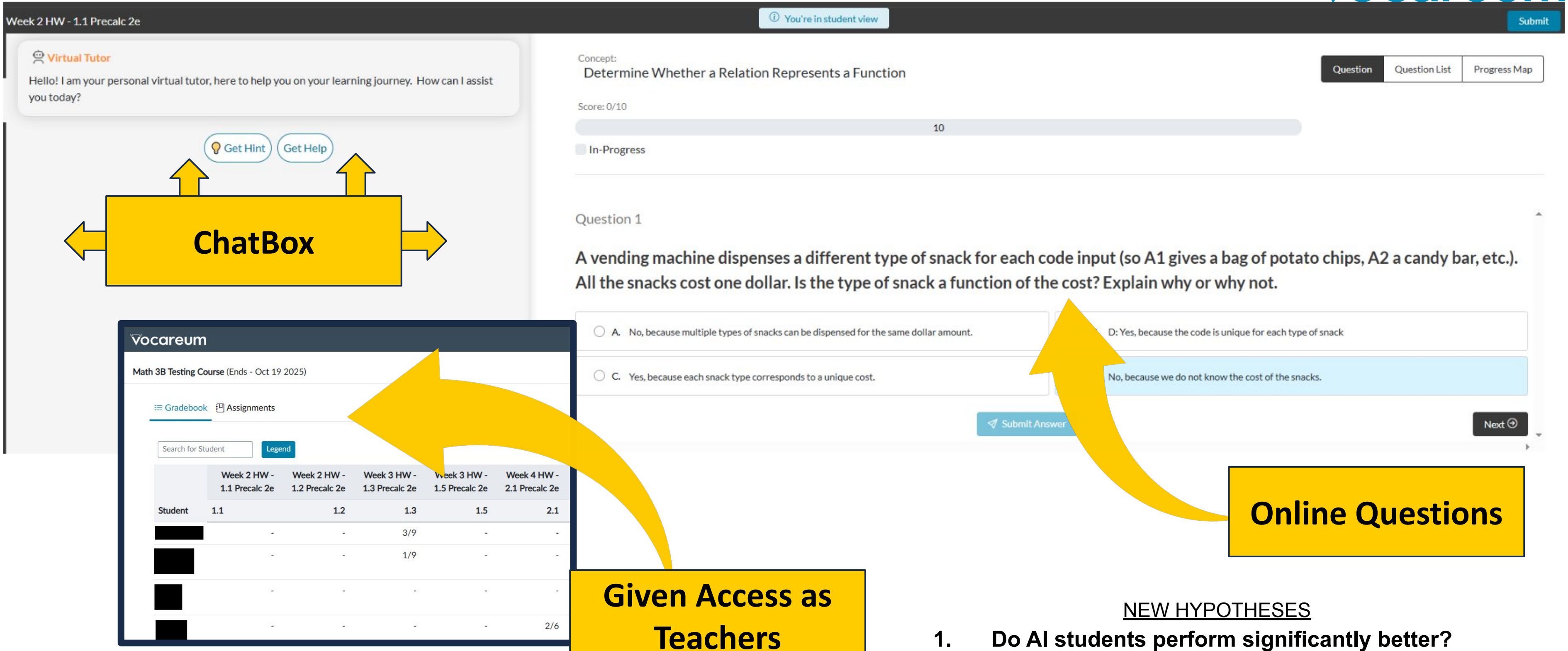
Unchanged

Key Summary Stats Here

Vocareum - Online Learning Platform

UC San Diego

vocareum



NEW HYPOTHESES

1. Do AI students perform significantly better?
2. Can development be done to comprehend usage?

AI Data - Conversations, Questions, etc.

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Conversations

Order	ChatID	VocareumID#	Timestamp	Date/Time	Message
120	2892	178742970	1743621501	4/2/2025 19:18	how would you rewrite a function?
121	2892	0	1743621509	4/2/2025 19:18	**_Key Idea:_**
122	2892	178742970	1743621621	4/2/2025 19:20	should we divide it
123	2892	0	1743621628	4/2/2025 19:20	**_Insightful Query!_**
124	2892	178742970	1743621772	4/2/2025 19:22	does it represent the variable?
125	2892	0	1743621778	4/2/2025 19:22	**_Great Thinking!_**
126	2893	178742970	1743622141	4/2/2025 19:29	can I combine like terms?
127	2893	0	1743622148	4/2/2025 19:29	**? Absolutely!**
128	2893	178742970	1743622254	4/2/2025 19:30	once I combine the p^2-p^2, they cancel out and I'm left with q, right?
129	2893	0	1743622261	4/2/2025 19:31	**?? Absolutely Correct!**
130	2894	178742970	1743622918	4/2/2025 19:41	would the answer be (0,-6)
131	2894	0	1743622931	4/2/2025 19:42	**_Let's Explore This Together!_**
132	2894	178742970	1743623128	4/2/2025 19:45	0
133	2894	0	1743623139	4/2/2025 19:45	**_Excellent Work!_**
134	2894	178742970	1743623194	4/2/2025 19:46	so would i put (0,-6) or (-6,0)
135	2894	0	1743623202	4/2/2025 19:46	**_Great Question!_**
136	2895	178742970	1743623310	4/2/2025 19:48	would i substitute the 4?
137	2895	0	1743623316	4/2/2025 19:48	**Absolutely!**

Conversation Length

ChatID	Student Questions	Total Messages (x2 to include AI Tutor response)
2892	3	6
2893	2	4
2894	3	6
2895	3	6
2896	7	14
2897	2	4
2898	5	10
2899	3	6
2900	3	6
2901	1	2
2902	1.5	3
2903	3	6
2904	2.5	5
2905	1	2
2906	1	2
2907	1	2
2908	1	2
2909	5	10

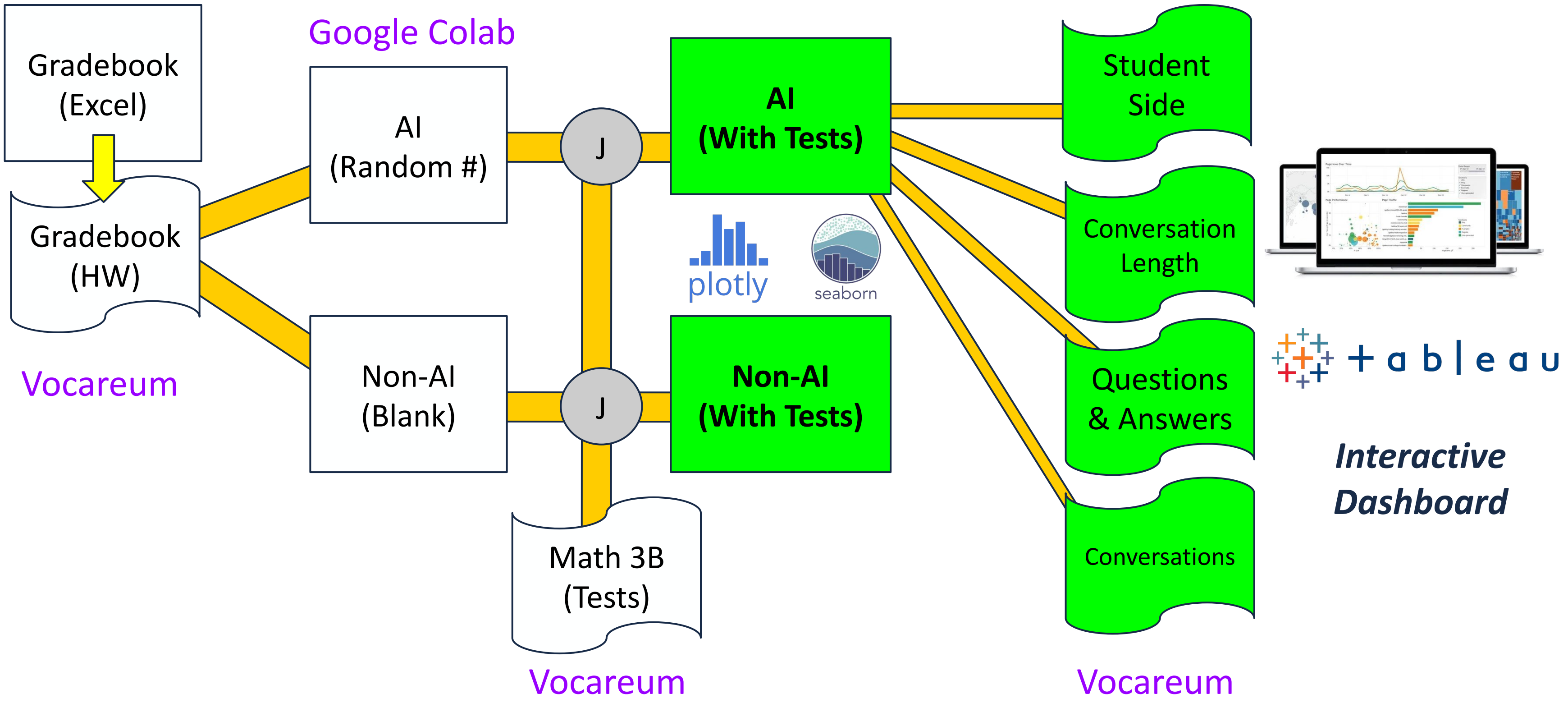
Questions/Answers

Order	attemptid	domainid	VocareumID#	conceptid	partid	questioni	response_	_resp	response_	assignment_ti	Date/Time	completion_time	quiz_type	attempt_r	ai_help_c	chat_conversation
143	8085	259	171841500	2611	3944273	4774	unanswere			1743124938	3/28/2025 1:22		0	0	0	[]
222	8164	259	171841500	2611	4078228	4774	correct			1743130206	3/28/2025 2:50	1743130223	0	0	0	[]
223	8165	259	171841500	2611	4078228	4776	correct	B		1743130225	3/28/2025 2:50	1743130228	0	0	0	[]
224	8166	259	171841500	2611	4078228	4785	correct	C		1743130230	3/28/2025 2:50	1743130235	0	0	0	[]
225	8167	259	171841500	2611	4078228	4786	correct	B		1743130236	3/28/2025 2:50	1743130243	0	0	0	[]
226	8168	259	171841500	2611	4078228	4787	correct	C		1743130246	3/28/2025 2:50	1743130252	0	0	0	[]
227	8169	259	171841500	2459	4078228	4769	correct	A		1743130253	3/28/2025 2:50	1743130291	0	0	0	[]
228	8170	259	171841500	2459	4078228	4770	correct	B		1743130293	3/28/2025 2:51	1743130298	0	0	0	[]
229	8171	259	171841500	2459	4078228	4771	correct	B		1743130299	3/28/2025 2:51	1743130304	0	0	0	[]
230	8172	259	171841500	2459	4078228	4773	correct	A		1743130306	3/28/2025 2:51	1743130315	0	0	0	[]
231	8173	259	171841500	2459	4078228	4845	correct	B		1743130318	3/28/2025 2:51	1743130325	0	0	0	[]
232	8174	259	171841500	2459	4078228	4846	skipped	-		1743130326	3/28/2025 2:52	1743130333	0	0	0	[]
233	8175	259	171841500	2460	4078228	4788	correct	B		1743130334	3/28/2025 2:52	1743130519	0	0	3	[2856]
234	8176	259	171841500	2460	4078228	4789	incorrect	B		1743130566	3/28/2025 2:56	1743130570	0	0	0	[]
236	8178	259	171841500	2460	4078228	4790	skipped	-		1743131296	3/28/2025 3:08	1743131552	0	0	1	[2857]
237	8179	259	171841500	2460	4078228	4791	skipped	-		1743131553	3/28/2025 3:12	1743131558	0	0	0	[]
238	8180	259	171841500	2460	4078228	4793	skipped	-		1743131558	3/28/2025 3:12	1743131565	0	0	0	[]
239	8181	259	171841500	2460	4078228	4789	skipped	-		1743131565	3/28/2025 3:12	1743131570	0	0	0	[]

Student Side

Order	ChatID	VocareumID#	Timestamp	Date/Time	Message	Key:
695	3064	178741845	1744002593	4/7/2025 5:09	ca[illegible] tell me step by step how to solve this?	"Step-by-step"
3494	3742	179357400	1745270071	4/21/2025 21:14	ca[illegible]te out the steps to solving this problem	"Step-by-step"
3963	3832	178744410	1745292653	4/22/2025 3:30	ca[illegible] show me how?	"Step-by-step"
2693	3531	179072820	1744738225	4/15/2025 17:30	Can i please get a step by step	"Step-by-step"
2607	3514	179072820	1744692494	4/15/2025 4:48	Can u give me step by step help	"Step-by-step"
2200	3434	178744410	1744678113	4/15/2025 0:48	can you give me a step by step	"Step-by-step"
3673	3776	178741845	1745281058	4/22/2025 0:17	Can you go step by step how to solve this please	"Step-by-step"
3629	3771	178741845	1745280072	4/22/2025 0:01	Can you go through it step by step	"Step-by-step"
3557	3760	178741845	1745277997	4/21/2025 23:26	Can you go through with me step by step	"Step-by-step"
3597	3763	178741845	1745278438	4/21/2025 23:33	Can you help me do this step by step	"Step-by-step"
3647	3774	178741845	1745280456	4/22/2025 0:07	Can you help me go by it step by step	"Step-by-step"
3711	3782	178741845	1745282455	4/22/2025 0:40	Can you help me step by step	"Step-by-step"
2749	3544	179072820	1744740027	4/15/2025 18:00	Further step by step	"Step-by-step"
2152	3426	178744410	1744677330	4/15/2025 0:35	give me a step by step explanation	"Step-by-step"
2056	3407	178743105	1744670769	4/14/2025 22:46	go step by step	"Step-by-step"
3164	3647	178741845	1745119669	4/20/2025 3:27	Go through it step by step and use the variables x y and m	"Step-by-step"
465	2993	178743240	1743813902	4/5/2025 0:45	please help me step by step	"Step-by-step"
2032	3402	178742970	1744669928	4/14/2025 22:32	step by step	"Step-by-step"
2615	3515	179072820	1744692972	4/15/2025 4:56	Step by step	"Step-by-step"

Data Pipeline



Data Cleaning - BEFORE

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Missing Values
Cannot 'compute'

Multiple Header Names
RandomUser, Random User #, Vocareum Random #, …

Long Names
"Week 4 HW - 2.1 Precalc 2e"

Scores by Grade/Max
Cannot 'compute'

Unused Rows
Dataframes ingest top 1

Student #	Vocareum Rand	Week 2 HW - 1.	Week 2 HW - 1.	Week 3 HW - 1.	Week 3 HW - 1.5
From Ko	From Voc	1.1	1.2	1.3	1.5
1	178739325	10/10	9/9	9/9	
2	#N/A	10/10	9/9	9/9	8/10
3	#N/A	10/10	9/9	Late	Late
4	178739370				

Both Types
Non-AI and AI

Dropped Tests
Shows '0' => *REGRADE*

Multiple CSVs
HWs vs. Tests

String "Late"
Cannot 'compute'

Dropped Class
No values anywhere

Data Cleaning - AFTER

UC San Diego

Easier Names
All same terminology
Fit onto charts

Student #	Vocareum ID	HW 1.1
4	178739370	10
5	179357400	10
6	178739730	10
7		
9		
1		

Calculated Scores
10% Tests, 20% MidTerm

HW 2.2	HW 2.3	HW 2.5	HW Score	HW GPA_Percent	HW Letter Grade	HW 1 Test
8	6	18	76	98.7	A+	42.8
8	7	18	77	100	A+	45
8	7	18	74	96.1	A	42
7	7	17	74	96.1	A	45
8	7	18	77	100	A+	45
7.831168831	6.945945946	18	74.8	97.1	[illegible]	38

CSV's Split
Now AI vs. Non-AI

Lates Handled
Now *'means'*

Numerical
Use functions/graphs

Letter Grades
Python Grade script

Data Frames

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140 Students
Non-AI: 39 AI: 64

Excels Provided: Class Tests (Math 3B Gradebook) and Homework (Anonymized Student Gradebook)

```
1 ai_df.head(3)
```

Student #	Vocareum ID	HW 1.1	HW 1.2	HW 1.3	HW 1.4	HW 2.1	HW 2.2	HW 2.3	HW 2.5	...	HW GPA_Percent	HW Letter Grade	HW 1 Test	HW 2 Test	HW 3 Test	HW 4 Test	Midterm 1	Test GPA Percent	Test Letter Grade	StudentMessageCount
4	178739370	10.0	9.0	9.0	10.0	6.0	8.0	6.0	18.0	...	98.7	A+	42.8	34.0	25.5	40.0	42.25	81.3	B-	35
5	179357400	10.0	9.0	9.0	10.0	6.0	8.0	7.0	18.0	...	100.0	A+	45.0	39.0	41.5	39.0	47.25	92.1	A-	19
6	178739730	10.0	9.0	8.0	8.0	6.0	8.0	7.0	18.0	...	96.1	A	42.0	40.5	42.5	38.0	12.25	50.6	F	11

```
1 none_df.head(3)
```

Student #	Vocareum ID	HW 1.1	HW 1.2	HW 1.3	HW 1.4	HW 2.1	HW 2.2	HW 2.3	HW 2.5	...	HW GPA_Percent	HW Letter Grade	HW 1 Test	HW 2 Test	HW 3 Test	HW 4 Test	Midterm 1	Test GPA Percent	Test Letter Grade	StudentMessageCount
8	NaN	10.0	9.0	9.0	10.0	6.0	8.0	7.000000	18.00	...	100.0	A+	45.0	45.0	44.5	43.0	40.75	87.4	B+	0
16	NaN	10.0	9.0	9.0	7.0	6.0	7.0	6.938776	17.16	...	93.6	A	44.0	43.0	43.0	13.0	41.00	80.0	B-	0
19	NaN	10.0	9.0	9.0	10.0	6.0	8.0	7.000000	18.00	...	100.0	A+	45.0	43.5	43.0	36.0	50.00	96.0	A	0

ai_df (top) and **none_df** (bottom) comparing similar metrics

- Also, <u>calculated</u> Scores, GPA %, and Letter Grades

Results - Dataframe Summaries

	Midterm 1	Test GPA Percent
count	39.000000	39.000000
mean	39.192308	80.356410
std	10.728170	17.140607
min	9.500000	23.800000
25%	35.250000	77.600000
50%	44.000000	86.800000
75%	47.000000	92.350000
max	50.000000	96.000000

Non-AI

	Midterm 1	Test GPA Percent	StudentMessageCount
count	64.000000	64.000000	64.000000
mean	38.921875	81.203125	11.281250
std	9.848644	13.421955	16.048012
min	5.250000	34.200000	1.000000
25%	34.187500	74.525000	2.000000
50%	40.250000	83.100000	4.500000
75%	47.250000	92.150000	15.000000
max	51.000000	98.400000	94.000000

AI Students

Min/Max

- AI students have higher Min (34 vs. 23) and higher Max (98 vs. 96)

Std Dev

- 13% AI vs. 17% Non-AI
- AI users do ‘similarly well’

25%, 50%, 75%

- Non-AI Users scored…*better*?

Mean

- Both scores basically the same (~39)

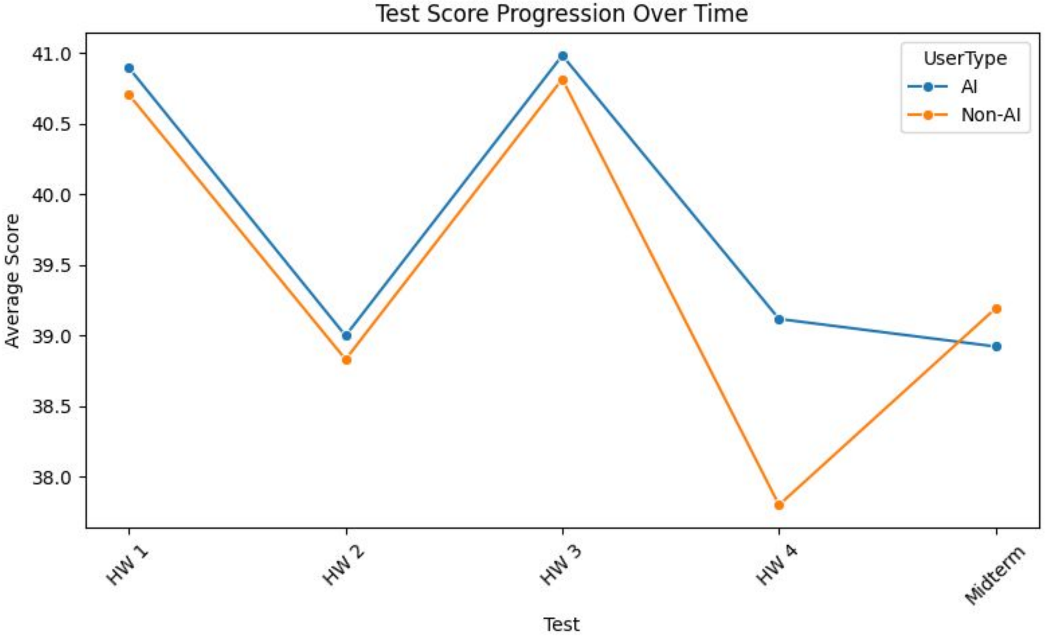
Results - Progress over Time

Tests performed without AI

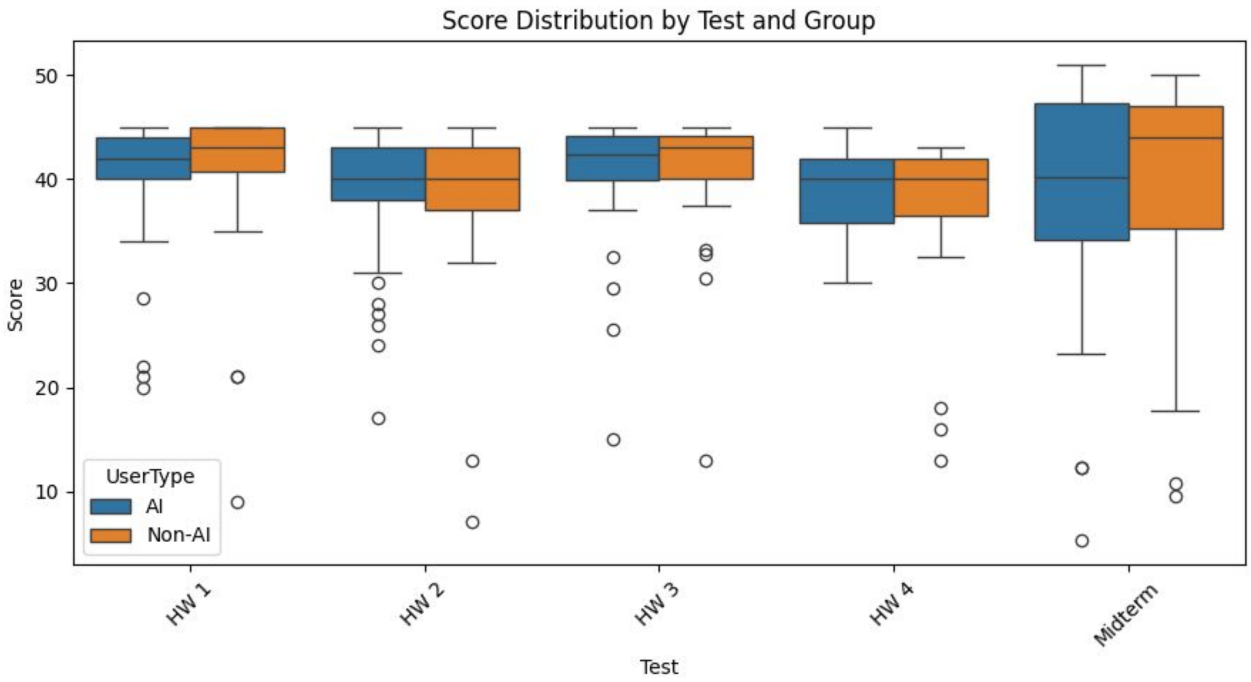
- Normal AI users do similar (HW1) or better (HW 4) on Tests leading up the the Midterm

- They actually performed worse during the Midterm - ***unexpected!***

"Why? Unknown. Could be Dependance on AI?"



Results - Deviations and Outliers



Scores and standard deviations tend to match
UP THROUGH the more-valuable Midterm

Results - Improvement Indicators

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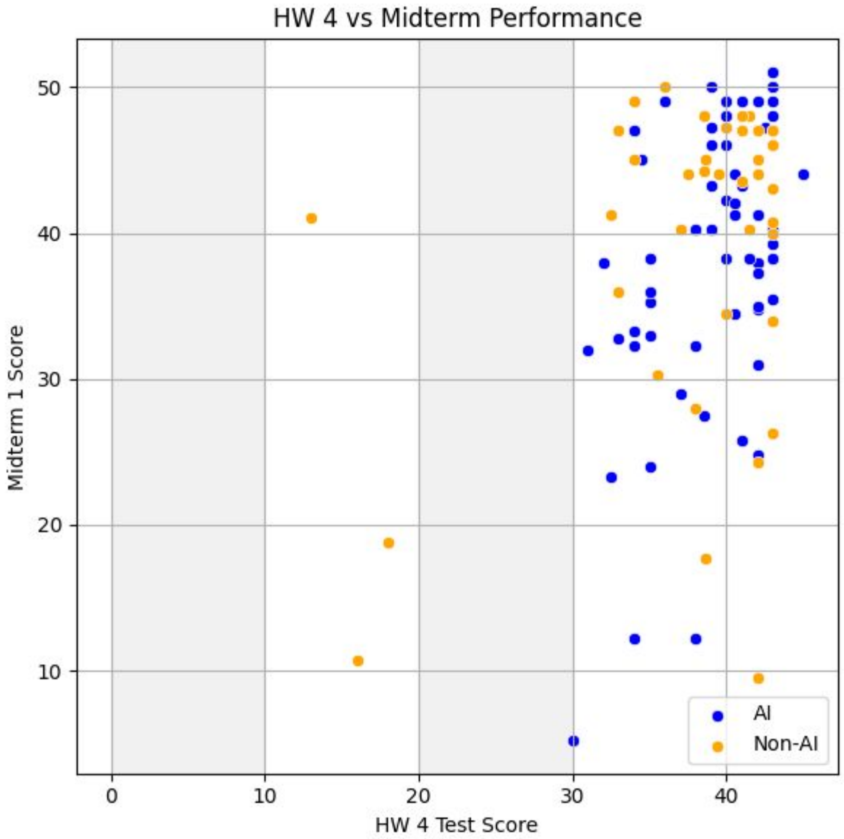
Progression from an earlier Test

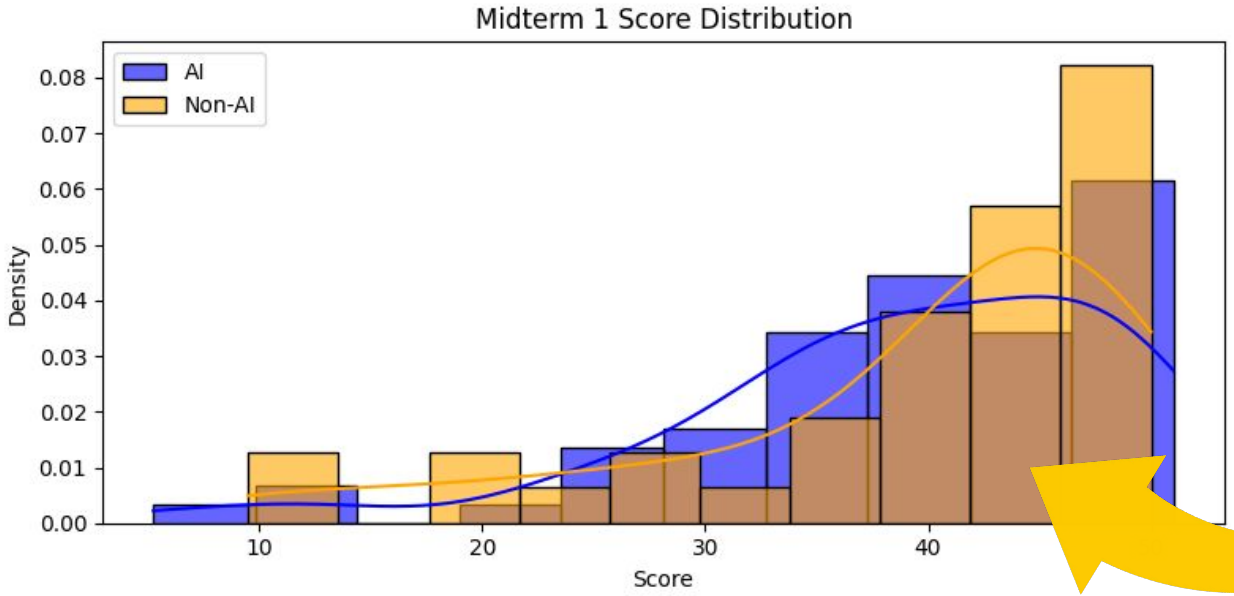
HW 4 Test to the Midterm

- Focus on that *last* HW Test
- 3 non-AI users in low HW Test range
- Low scoring Midterms have various
- Catch them ***early***:
    - Alert ALL the <u>outliers</u>
    - View with their AI interaction
    - Motivate for higher AI usage?

In general, most did as expected
(in upper right quadrant)



Results - Score Distributions



Note the *dip* at AI 'high range'
What would cause this? Their usage stats? Need to dig deeper…

Shows the unexpected result of non-AI students' high comparative scores

Nerd Alert: Using scipy ttest_ind, even for most extreme case (> 42 pts):

p-score = 0.12 → Not under 0.05 threshold → No statistical difference

Results - Outcomes by AI Usage

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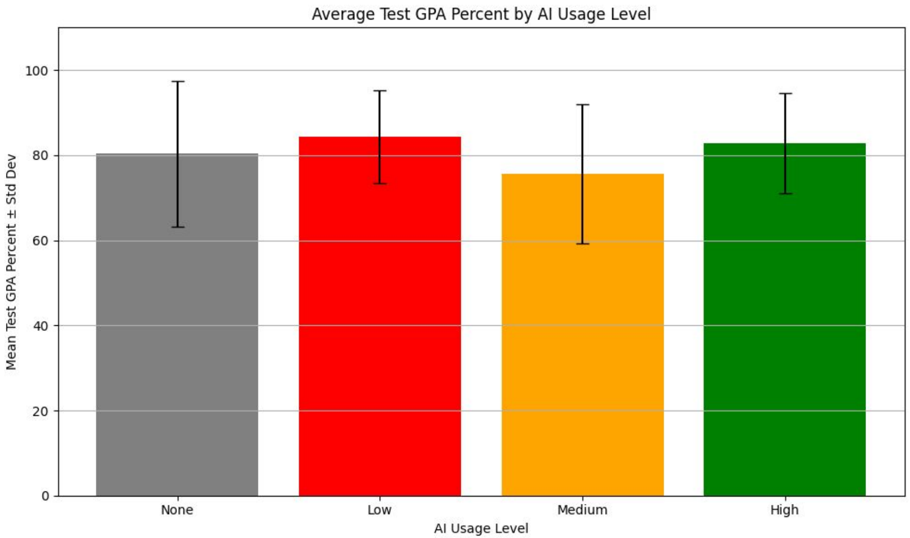
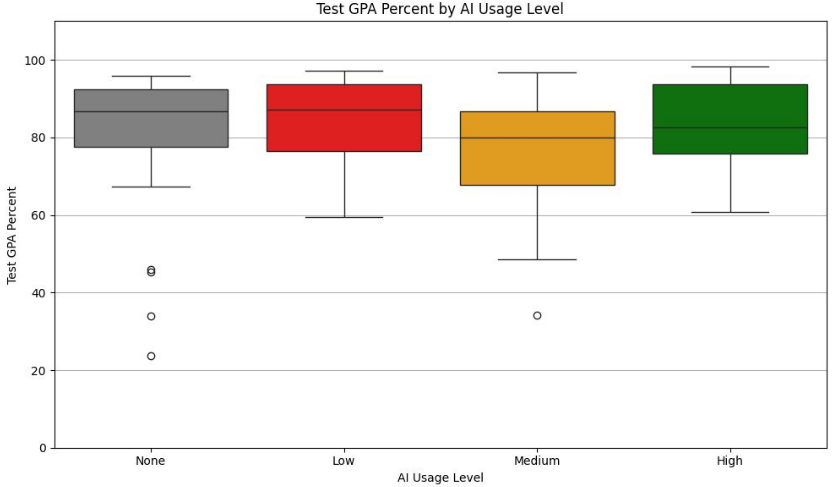
Bar Chart: **Means**



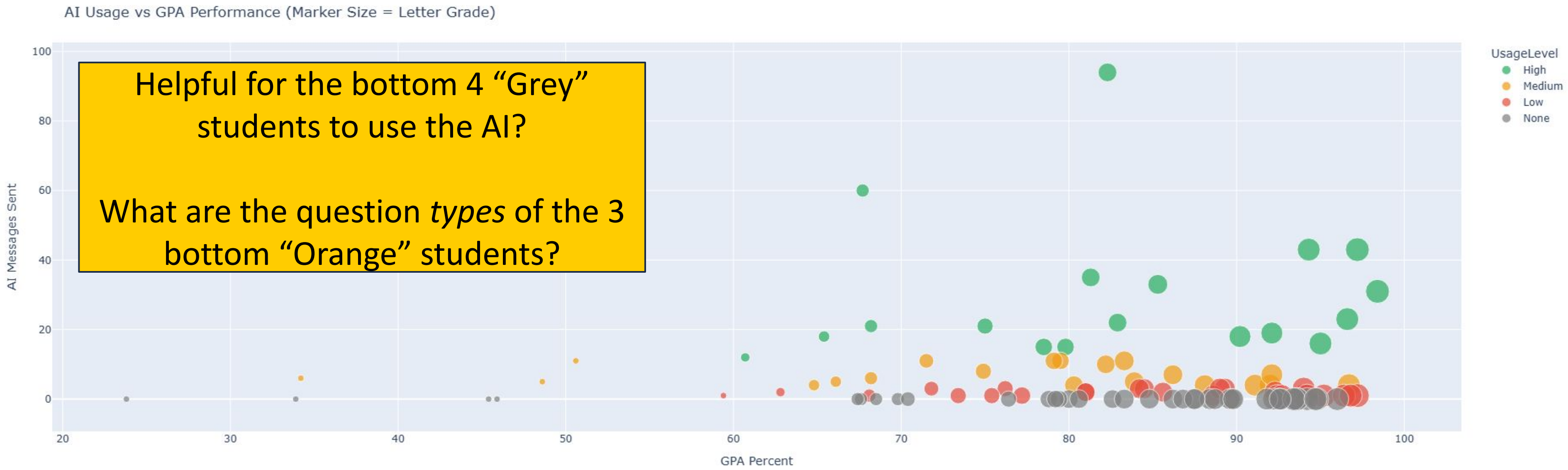
Box Plot: **Median**



AI Msgs: **Grey** = No AI use **Red** = Low, <3 **Orange** = Medium, 3–12 **Green** = High AI use, 12–100

- Shows minimal differences across usage levels.
- Medium users show noticeable difference, but within range

Results - Scores by Usage



Recommended to teachers for instant comparison of a <u>single</u> student to entire class

Scatter provides a more granular look.

- Dots: Isolate each student
- Color: Isolate usage group
- Size: Letter Grade

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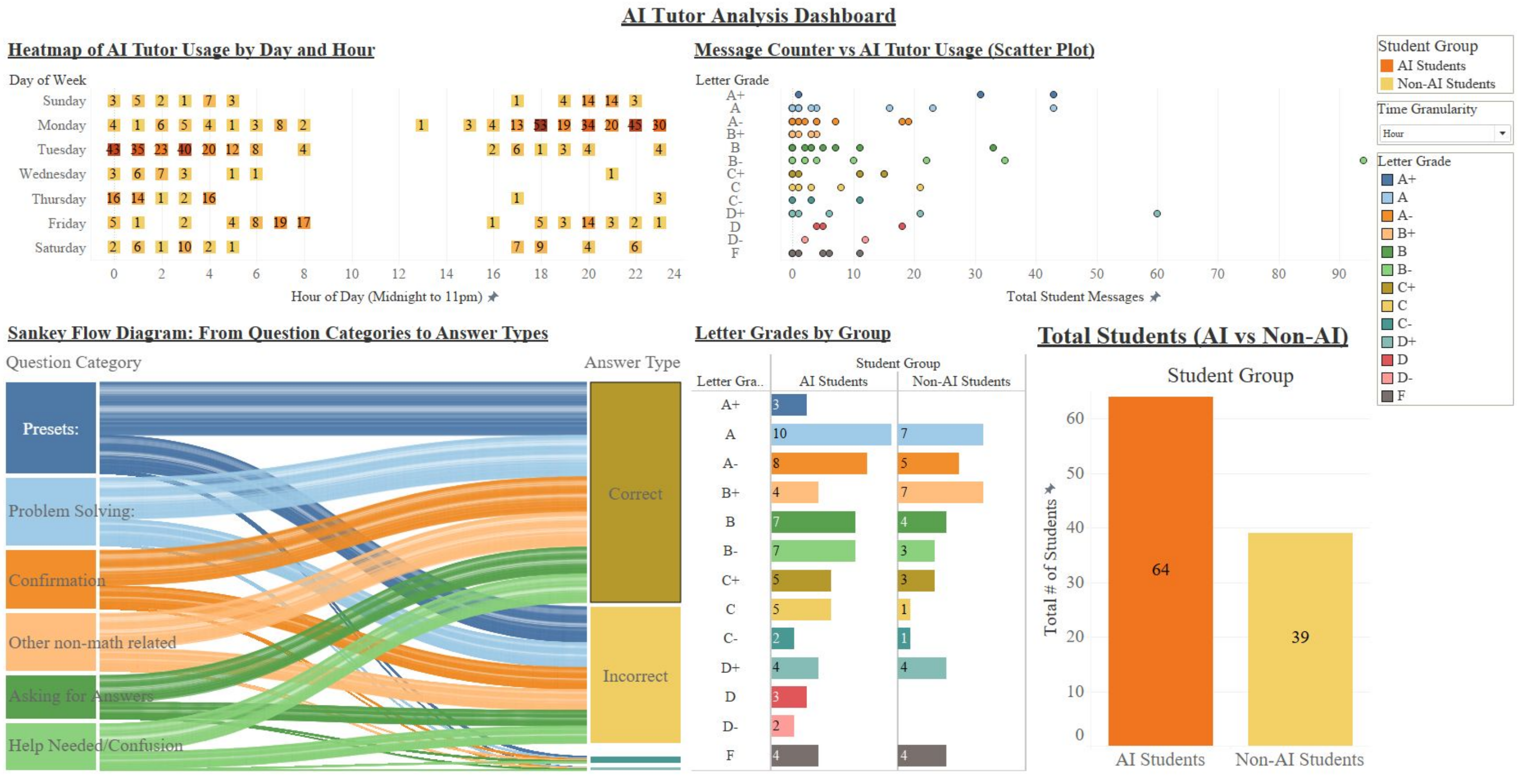
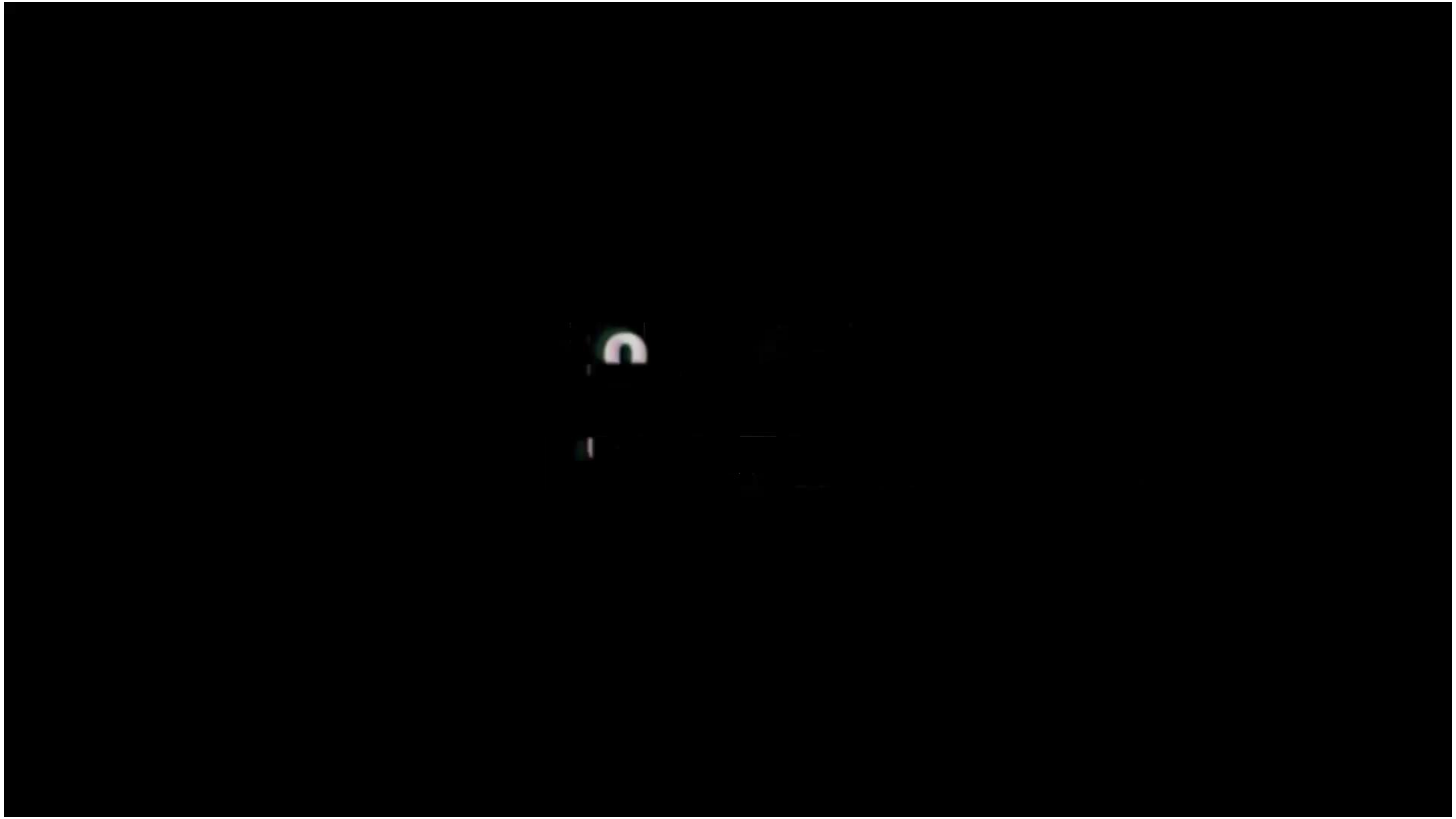
DEMO - AI Tutor Analysis Dashboard



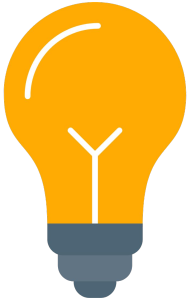
Interesting insights found:

- more students used the tutor
- most correct answers
- hours and days AI tutor was used

DEMO - AI Tutor Analysis Dashboard



Value of Results



1) **Similarities between AI and non-AI students' scores show that AI can be helpful, but also can be relied on too much**

2) **Most students perform well in both groups but there are some that need improvement**

3) **<u>MOST IMPORTANT</u>: Vocareum and the UC San Diego teams now have a functional pipeline that can be used for any course (e.g. CSS-1). They will be able to visualize the data from each course to find out:**
    - **how many students used the AI tutor**
    - **the distribution of Message Counts to Letter Grades**
    - **how many "correct", "incorrect", etc answer types for each category**
    - **the number of students who use the AI tutor on which day and at which hour**

Conclusions - Hypothesis #1

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Data shows limited improvement in student scores using Vocareum

Key Insights from Final Datasets:

- AI group has 64 students, Non-AI has 39 (of 140 in set)
- AI group - Higher scores for HW Tests but not Midterm

This negates our prior hypothesis that AI users would perform better.

- The **describe()** feature shows the Midterm scores are basically the same (~39) for both types of users.
- The fact that the AI users performed consistently better on all the **homework tests**, sometimes substantially, confuses the issue.
- This combined with the **graph** showing lower and fewer high-scores on the Midterm means that AI users do not statistically have an advantage. This was verified with a **p-test**.

This outcome can be related to a number of factors. Initial considerations are that the AI users were too dependent on the AI for answers, or had less preparation due to not considering hard practice homework questions. Additional data exploration into how the AI system was used using additional information received from CSVs obtained may reveal the root cause.

Conclusions - Hypothesis #2

UC San Diego

Successfully developed end-to-end pipeline culminating in interactive graphical dashboard

Key Insights from Final Dashboard:

1. **Bar Charts**: Categorical & Numerical comparisons. Letter Grades show similarity for both groups.
2. **Sankey** - Categories have highest count of correct answers, then incorrect, and so on.
3. **Heatmap** - Higher AI usage in the evening of Monday & morning of Tuesday than rest of week.
4. **Scatter** - Most students used AI less and received decent grades (A - C-).

The interactivity allows singling out students, focusing on AI interaction, focusing on timeframes, and separating usage statistics into specific sections

Can be used with multiple class majors, student sizes, and varied assignments

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